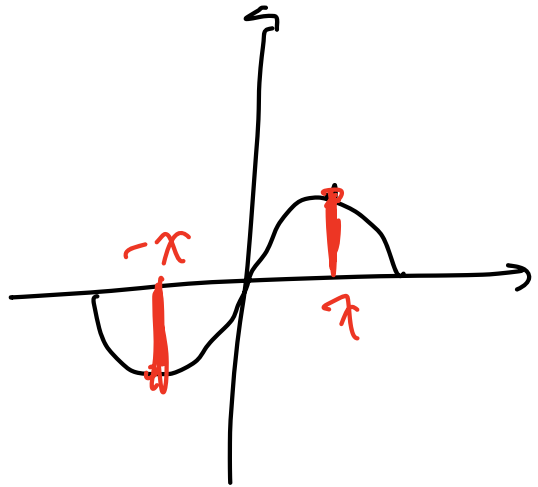


奇偶性 or "配对"

$f(x)$ 奇函数

$$f(x) + f(-x) = 0$$

$$\int_{-a}^a f(x) dx = 0$$



$$\int_{-1}^1 \boxed{\frac{1}{1+e^x}} dx$$

$f(x)$

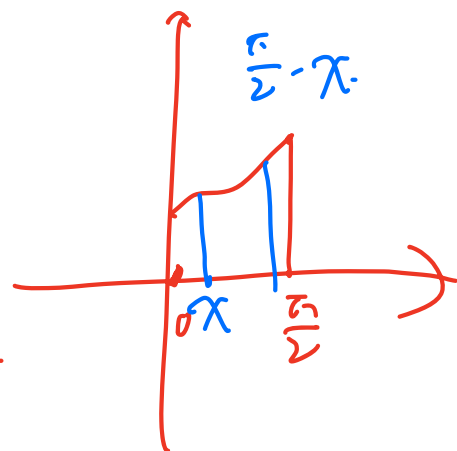
$$\begin{aligned} f(x) + f(-x) &= \frac{1}{1+e^x} + \frac{1}{1+\frac{1}{e^x}} \\ &= \frac{1}{1+e^x} + \frac{e^x}{e^x+1} \\ &= 1 \end{aligned}$$

$$2 \int_{-1}^1 \frac{1}{1+e^x} dx = [x(1-(-1))] = 2$$

$$\Rightarrow \int_{-1}^1 \frac{1}{1+e^x} dx = 1.$$

$$\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x} dx$$

$$\frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}}$$



$$f(x) = \frac{\sin x}{\sin x + \cos x}$$

$$f(x) + f\left(\frac{\pi}{2} - x\right) = \frac{\sin x}{\sin x + \cos x} + \frac{\sin\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)}$$

$$= \frac{e^{\sin x}}{e^{\sin x} + e^{\cos x}} + \frac{e^{\cos x}}{e^{\sin x} + e^{\cos x}} + \frac{\sin\left(\frac{\pi}{2} - x\right)}{\sin\left(\frac{\pi}{2} - x\right) + \cos\left(\frac{\pi}{2} - x\right)}$$

$$= 1 = \frac{\sin x}{\sin x + \cos x} + \frac{\cos x}{\sin x + \cos x}$$

$$= 1$$

$$\int_0^{\frac{\pi}{2}} f(x) dx = \frac{1 \times \frac{\pi}{2}}{2}$$

$$= \frac{\pi}{4}$$

$$\int_{-1}^1 \underline{x^2 (\sin x + x^2)} dx$$

$$\underbrace{x^2 \sin x}_{\text{odd}} + \underline{x^4}$$

$$= \int_{-1}^1 x^4 dx = \left. \frac{1}{5} x^5 \right|_{-1}^1 = \frac{2}{5}$$

周期性.

$$\int_0^{2025} |\sin x| dx$$

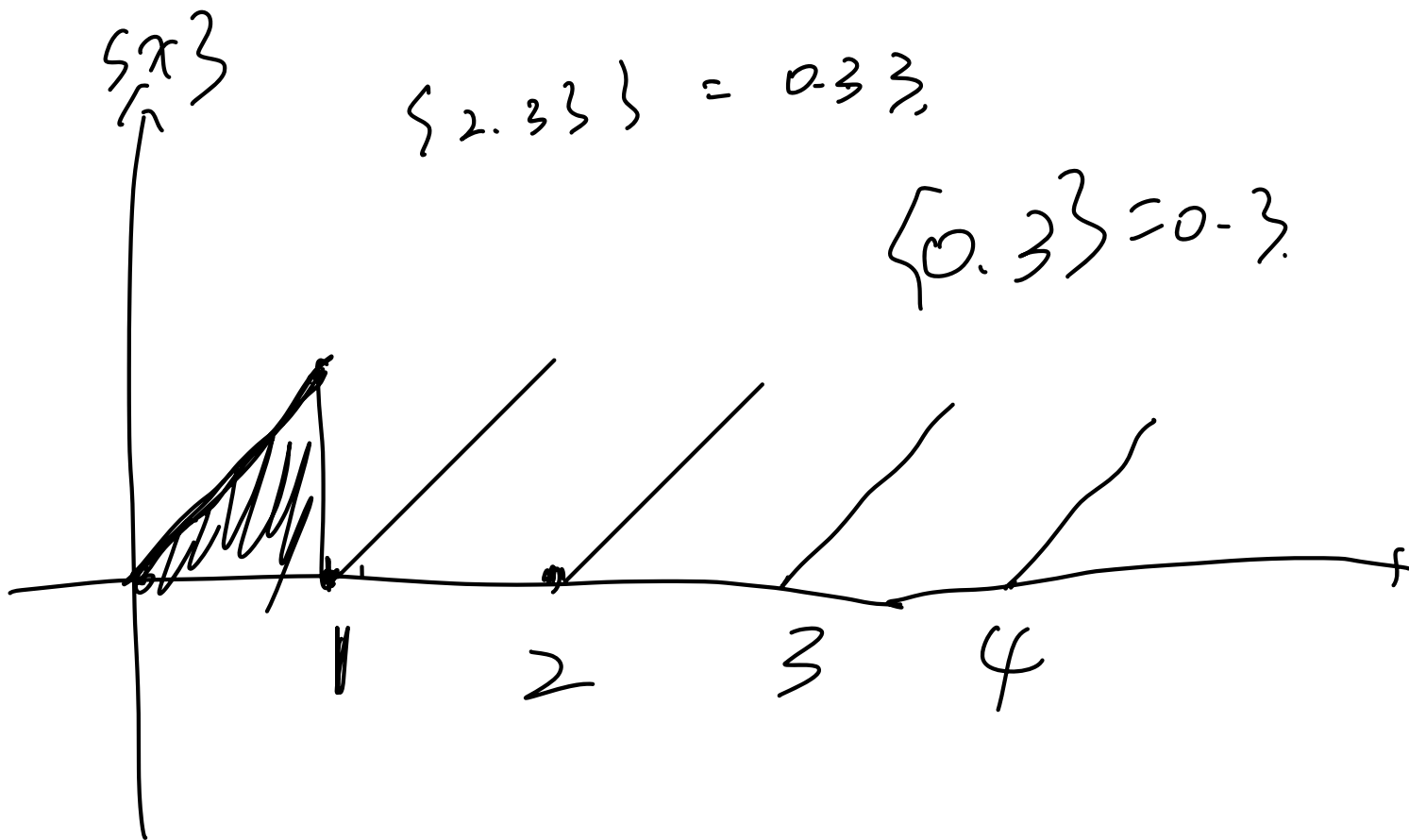
$$\int_0^{2025} \{x\} dx$$

$\{x\}$ 代表 x 的小数部分

$$\{4.7\} = 0.7.$$

$$\{2.33\} = 0.33,$$

$$\{0.3\} = 0.3.$$



$$= 2025 \int_0^1 \{x\} dx$$

$$= 2025 \int_0^1 x dx = \frac{2025}{2}$$

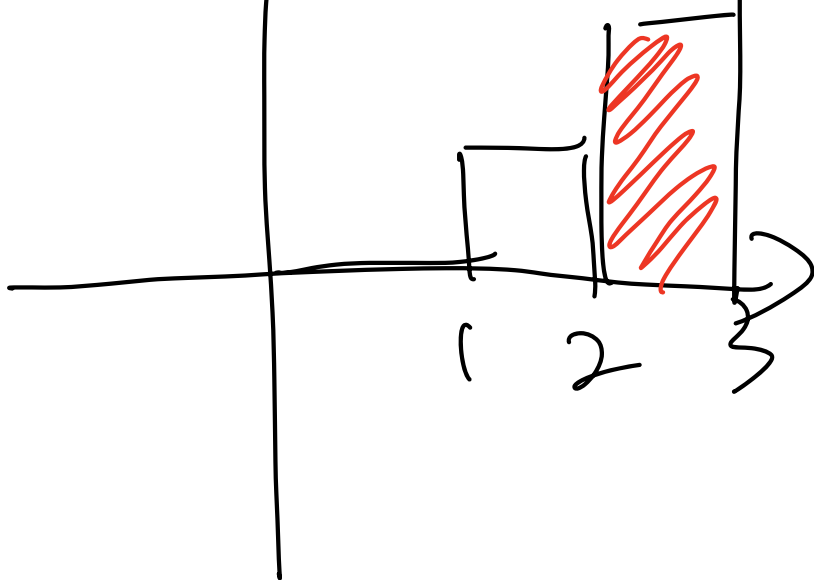
$$\int_0^{2025} \{x\} dx$$

$\{x\}$ 是不超过 x
的最大整数

$$\{1.1\} = 1$$

$$\{2.3\} = 2$$





$$\int_0^{2025} \lfloor x \rfloor dx$$

$$= \int_0^1 \underbrace{\lfloor x \rfloor}_{0} dx + \int_1^2 \underbrace{\lfloor x \rfloor}_{x-1} dx + \int_2^3 \underbrace{\lfloor x \rfloor}_2 dx$$

$$+ \underbrace{\int_{2024}^{2025} \lfloor x \rfloor dx}_{2024}$$

$$= 0 + 1 + 2 + 3 + \dots + 2024$$

$$= \frac{(1+2024) \times 2024}{2}$$

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x \sin x \, dx \quad \sin x \cdot \sin x$$

$$\sin x \sin x + \sin (\sin (-x))$$

$$= - - - + \sin (-\sin x)$$

$$= \sin x \sin x + (-\sin x \sin x) = 0$$

$$\sin (\cos (-x))$$

$$= \sin (\cos x)$$

$$\lim_{h \rightarrow 0} \frac{\sin (x+h)^2 - \sin x^2}{h}$$

$$\underline{f(x) = \sin x^2 \rightarrow 2x \cos x^2}$$

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = f'(x)$$

L'Hôsp

$$\lim_{h \rightarrow 0} \frac{2(x+h) \cdot \cos(x+h)^2 - 0}{1}$$

$$\lim_{t \rightarrow 0} \frac{\tan\left(\frac{\pi(1+t)}{2}\right) \cdot \ln(1+t)}{\infty}$$

比值

$$\checkmark \quad \lim_{t \rightarrow 0} \tan\left(\frac{\pi(1+t)}{2}\right) \cdot t$$

$$\ln(1+t) \sim t \quad (t \rightarrow 0)$$

$$= \lim_{t \rightarrow 0} \frac{\tan\left(\frac{\pi(1-t)}{2}\right)}{\frac{1}{t}} \cdot \frac{t}{\frac{1}{\tan\left(\frac{\pi}{2}(1-t)\right)}}$$

$$= \lim_{t \rightarrow 0} \frac{\frac{1}{\cos^2\left(\frac{\pi(1-t)}{2}\right)} \cdot \left(\frac{\pi}{2}(1-t)\right)'}{-\frac{1}{t^2}}$$

$$= \lim_{t \rightarrow 0} \frac{\frac{1}{\cos^2\left(\frac{\pi}{2}(1-t)\right)} \cdot \left(-\frac{\pi}{2}\right)}{-\frac{1}{t^2}}$$

$$= \lim_{t \rightarrow 0} \frac{\pi}{2} \cdot \frac{t^2}{\cos^2\left(\frac{\pi}{2}(1-t)\right)}$$

$$= \lim_{t \rightarrow 0} \frac{\pi}{2} \cdot \frac{t^2}{\cos^2\left(\frac{\pi}{2} - \frac{\pi}{2}t\right)}$$

$$\begin{aligned} t &\rightarrow 0 \\ 1-t &\rightarrow 1 \\ \frac{\pi}{2}(1-t) &\rightarrow \frac{\pi}{2} \end{aligned}$$

$$\cos\left(\frac{\pi}{2}(1-t)\right) \rightarrow 0$$

$$= \lim_{t \rightarrow 0} \frac{\pi}{2} \cdot \frac{t^2}{\sin^2\left(\frac{\pi}{2}t\right)}$$

$$= \lim_{t \rightarrow 0} \frac{\frac{\pi}{2}}{2t} \frac{2t}{2 \cos\left(\frac{\pi}{2}(1-t)\right) \cdot (-\sin\left(\frac{\pi}{2}(1-t)\right)) \cdot \left(\frac{\pi}{2}(1-t)\right)'}$$

11

11 . . .

$$\lim_{t \rightarrow 0} \tan\left(\frac{\pi}{2}(1-t)\right) \cdot t$$

$$= \lim_{t \rightarrow 0} \frac{\overset{1}{\sin\left(\frac{\pi}{2} - \frac{\pi}{2}t\right)}}{\underset{0}{\cos\left(\frac{\pi}{2} - \frac{\pi}{2}t\right)}} \cdot t^0$$

$$= \lim_{t \rightarrow 0} \frac{t}{\cos\left(\frac{\pi}{2} - \frac{\pi}{2}t\right)}$$